19 Solve $22 < \frac{m^2 + 7}{4} < 32$

Show all your working.

Multiply all three parts by 4:

$88 < m^2 + 7 < 128$

(−7) (−7) (−7)

$81 < m^2 < 121$

1 mark
Begin rearranging to isolate m^2

1 mark
Both bounds on m^2 correct

$m^2 = 81$ gives $m = -9$ or 9.

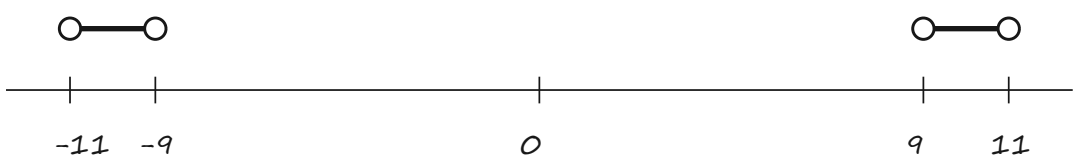
$m^2 = 121$ gives $m = -11$ or 11.

1 mark
Find all four critical values

$81 < m^2 < 121$ means $9 < |m| < 11$.

m is between 9 and 11, or between −11 and −9.

1 mark
Select the intervals



$-11 < m < -9$ or $9 < m < 11$

All endpoints are excluded: the inequalities are strict.

1 mark
Give both strict intervals

(Total for Question 19 is 5 marks)